CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B.Tech. Examination (CL)

December 2013

CL308 Geotechnical Engineering-II

Date: 05.12.2013, Thursday **Time: 01:30p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Enlist the methods of borings. Explain any two in detail. [05]

(b) What are the objectives of soil exploration? [02]

Q – 2 (a) **Attempt Any Two:** [10]

(i) List out the forces acting on the well foundation. Explain any two in detail.

(ii) Draw neat sketch of well foundation which shows different components of well. Write detail note on grip length.

(iii) How many shapes of well foundation can be prepared? Draw any four of them & write where they can be used.

(c) Give the differences between active earth pressure and passive earth pressure. [04]

Q - 3 (a) A retaining wall with vertical back of 9 m, high supports a horizontal backfill surface with c' = 0, Φ' = 33°. Calculate total active thrust on the wall, when water table is below the base of wall. Take $\gamma = 16\ kN/m^2$. [05]

(b) **Attempt Any Three:** [09]

(i) State the assumptions made in Boussinesq's theory on stress distribution.

(ii) A concentrated load of 100 kN acts on the surface of soil. Plot the variation of horizontal stress increment due to load on vertical plane at a distance of 3m up to a depth of 2 m (vertical intervals = 0.50m). Also plot the graph.

Depth 'z'	0.5	1.0	1.5	2.0
Influence factor 'I_B'	0.0085	0.0844	0.1904	0.2733

(iii) Explain in detail the equivalent point load method for calculating vertical stress.

(iv) Explain how to use Newmark's Influence chart.

SECTION – II

Q - 4 (a) What is stability number? Explain its uses in detail. [04]

(b) List various methods of improving stability of slopes. [03]

Q - 5 (a) An embankment of height 11 m is made of C-Φ soil having C = 15 kPa, Φ = 32°& γ=20 kN/m^3. The slope of embankment is 1.5H: 1V. The directional angles are α=26° & β=35°. Determine the factor of safety for the slip surface using Swedish method of slices. [05]

(b) A 12m long, 300mm diameter pile is driven in a uniform deposit of sand (Ø=40°). The water table is at a great depth and is not likely to rise. The average dry unit weight of sand is 18kN/m^3. Using Nq=135 calculate the safe load capacity of the pile with F.O.S. =2.5. Take coefficient of earth pressure 'K' = 1.0 and value of δ = 30°. [06]

OR

(b) In a 16 pile group, the pile diameter is 0.4 m and center to center spacing of piles in the square group is 1.5 m. If Cu=50 kN/m², determine whether the failure would occur as block failure or when the piles act individually. Neglect bearing at the tip of the pile. All piles are 12 m long. Take m=0.7 for shear mobilization around each pile. Also determine the safe load on this group [06]

(c) Explain with sketch group action of piles. [03]

Q - 6 (a) Explain in detail, any four factors influencing the selection of piles. [02]

(b) Compute the allowable bearing capacity of a square footing of 2 m size resting on dense sand of unit weight 20 kN/m^3. The depth of foundation is 1 m & the site is subject to flooding. The bearing capacity factors are: Nc = 55, Nq = 38, Nγ = 45. [07]

(c) What is meant by bearing capacity of soil? How will you determine in the field? Describe the procedure bringing out its limitations. [05]

OR

(b) Discuss the various factors that affect the bearing capacity of a shallow footing. Write brief critical notes on settlement of foundations. How do you ascertain whether a foundation soil is likely to fail in local shear or general shear? [07]

(c) Elaborate the word footing and foundation. Explain the factors that affect the selection of type of foundation. [05]

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